Presentation prep sheet

DTMS_Paper_Presentation_Short.pptx | 13 slides | 12 minutes | about 55 seconds per slide

The one-sentence version of the whole talk. We measured whether common database tuning advice actually works, on two databases and 23,480 measurements, and all four pieces of advice we tested make things worse under conditions we can now name.

The one thing to keep straight. Slides 1 to 7 compare two databases, PostgreSQL and MariaDB. Slides 8, 9 and 10 are about PostgreSQL only. The slide says so at the top. If you remember nothing else, remember that, because it is the question you are most likely to be asked.

PART 1 | SETUP (slides 1-5, about 4 minutes)

SLIDE 1 Title

What is on it. Title, your name, supervisor, and the scale of the study.

What it means. Nothing to explain.

Say this. "This is a study of how databases decide which index to use, and whether the advice people give about it actually works."

SLIDE 2 Motivation: a decision nobody sees

What is on it. The one decision every query needs, and the four pieces of advice we tested.

What it means. Every query needs one choice made for it: scan the whole table, or use an index, and if so which one. The user never sees it, but it can make the query a hundred times slower.

Say this. "Because this decision is invisible, a lot of advice has grown around it. It sounds sensible, and almost none of it has been measured. That is what I did."

SLIDE 3 Why two databases

What is on it. The reason the study uses PostgreSQL and MariaDB rather than one.

What it means. If you test one database and find a problem, you cannot tell whether you found something true about databases in general or just a bug in that one product. Two independently written databases let you separate those.

Say this. "If both databases fail the same way, it is a design problem. If only one fails, it is that product's choice. Every result later is labelled one way or the other."

SLIDE 4 The two measures

What is on it. Definitions of regret and q-error.

What it means. Regret = how many times slower the plan it chose was, compared to the best plan it could have chosen. Regret of 1.0 means it did the best it could. q-error = how badly it guessed the number of rows. 1.0 is a perfect guess.

Say this. "I keep these two apart on purpose, because the main finding is that they come apart: the guess was right and the choice was still wrong."

If asked: **Why not just measure speed?** Because then a slow database looks bad even when it made the right choice. Regret measures the decision, not the engine.

SLIDE 5 The experiment

What is on it. The table of how many measurements at each table size.

What it means. 23,480 measurements. Both databases cover the same seven table sizes, so the comparison is fair.

Say this. "I generated the data instead of using a standard benchmark, because that way I know the true row count for every query. So I can say the database guessed wrong, instead of guessing along with it."

PART 2 | COMPARING THE TWO DATABASES (slides 6-7, about 2 minutes)

SLIDE 6 Selection quality: PostgreSQL 2.3%, MariaDB 66.0%

What is on it. A two-row table comparing how often each database chose badly.

What it means. PostgreSQL picks the best available plan 97.7% of the time. MariaDB gets it wrong on two thirds of queries, but its mistakes are milder (median 1.34x versus PostgreSQL's worst of 1.39x).

Say this. "Read the last two columns together. PostgreSQL is almost always right. MariaDB is usually wrong, but only by a little. Two different kinds of failure, not one database being better at everything."

If asked: **Is the 2.3% with or without BRIN?** Without. That is exactly what slide 8 is about, so say "I am coming to that in two slides."

SLIDE 7 Why they differ: the plans available

What is on it. A count of what each database actually chose: bitmap scan, index lookup, or full table scan.

What it means. PostgreSQL uses a bitmap scan 346 times; MariaDB 7 times. MariaDB gave up and scanned the whole table 66 times; PostgreSQL 9 times.

Say this. "A bitmap scan sorts the matching rows into disk order before fetching them. That puts a ceiling on how bad a mistake can be, because even a wrong choice still reads the disk in order. MariaDB has no equivalent, so it has no ceiling."

If asked: **So MariaDB is just worse?** Not exactly. It is wrong more often but more mildly. Slide 11 shows that reverses as tables grow.

PART 3 | THE MAIN FINDING, IN THREE STEPS (slides 8-10, about 3 minutes)

These three slides are one story, in order: **what happened** → **what it was not** → **what it actually was**. All three are PostgreSQL only.

SLIDE 8 Adding one more index broke it

What is on it. Two rows. Without a BRIN index: 2.3% bad choices. With one: 73.3%.

What it means. Same database, same data, same queries. The only thing that changed is whether a BRIN index exists on the column next to the B-tree. People add BRIN indexes because they are tiny and look free.

Say this. "On slide 6 I said PostgreSQL chooses well, 2.3% bad. That was without a BRIN index. Watch what happens when I add one: 73.3%. This is not a contradiction of slide 6, it is what adding one supposedly free index does to it."

If asked: Is this both databases? No, PostgreSQL only. MariaDB does not have BRIN indexes at all, so it cannot make this mistake. That is how we know it is a PostgreSQL costing defect and not something unavoidable.

SLIDE 9 It was not bad estimates

What is on it. 98.2% of those 113 bad choices had the row count essentially right.

What it means. The usual explanation for a bad query plan is that the database misjudged how many rows would come back. Here it did not. It knew the row count, it knew both indexes existed, and it still chose the slower one.

Say this. "This rules out the usual fix too. Running ANALYZE or adding statistics cannot help, because you cannot improve a number that was already correct. So if the estimates were right, what did it get wrong?"

If asked: Where does 113 come from? It is the bad choices from the previous slide: PostgreSQL at 1,000,000 rows. 110 of the 113 come from the with-BRIN case, which is why these two slides belong together.

SLIDE 10 It compares the wrong thing

What is on it. BRIN costs 12.13 to read and takes 65.95 ms. B-tree costs 213.35 to read and takes 0.34 ms.

What it means. When two indexes can answer the same query, PostgreSQL keeps whichever is cheaper to read and throws the other away. But reading the index is only half the job; the query then has to fetch the actual rows, and that part is not counted. BRIN is tiny so it always wins, and the index thrown away is the fast one.

Say this. "The index that costs 17 times less to read is the one that runs 194 times slower. It is comparing the wrong quantity."

If asked: How do you know this is the cause? I read the PostgreSQL source. The function is choose_bitmap_and in indxpath.c, and it compares index-read cost only. It is on the slide.

PART 4 | THE WIDER PATTERN (slides 11-13, about 3 minutes)

SLIDE 11 The two databases fail in opposite directions

What is on it. Two charts: how often each database errs, and how badly, as tables grow.

What it means. As tables get bigger, PostgreSQL is wrong more often (0% up to 36%) but never by more than 3.7x. MariaDB is wrong less often (53% down to 12%) but its worst case climbs to 30.9x. They cross over.

Say this. "A database that is wrong often but never badly is easier to run than one wrong rarely but catastrophically. If you summarise either with a single number you lose exactly this."

SLIDE 12 Four practices, all four backfire

What is on it. Four rows: what each piece of advice promises, and what we measured.

What it means. In three of the four, the advice does exactly what it claims and the query still gets slower. Extended statistics really do fix the estimate by 108x. The plan gets worse anyway.

Say this. "That is the pattern of the whole study. The advice improves the thing it promises to improve, and that thing is not what determines how fast the query runs."

SLIDE 13 What to do, and what not to do

What is on it. Four things to do on the left, four not to do on the right, each with the number behind it.

What it means. This is the practical payoff. Every 'do not' has a measured number attached so it is a finding, not an opinion.

Say this. "All four were reasonable advice. None of it shipped with the conditions under which it stops being true. That is what this study adds: the conditions."

The four numbers you must not mix up

2.3% PostgreSQL, WITHOUT a BRIN index. Its normal, good behaviour. Slides 6 and 8.

73.3% PostgreSQL, WITH a BRIN index added. Same data, same queries. Slide 8.

98.2% Of the 113 cases where PostgreSQL chose badly, the share that had a correct row estimate. Slide 9.

194x How much slower one query got, purely because a BRIN index existed. Slide 10.

66.0% MariaDB's bad-choice rate. Nothing to do with BRIN; MariaDB has no BRIN. Slide 6.

Why 2.3% and 73.3% are not a contradiction

They are the same database on the same data. The only difference is which indexes existed when the query ran. Without BRIN it chooses well. Add a BRIN index beside the B-tree and it chooses badly, because of the costing defect on slide 10. If someone thinks you contradicted yourself, that one sentence resolves it.

Three questions worth rehearsing

Why did you not use a standard benchmark like TPC-H?

Because a fixed dataset fixes the skew, the correlation and the clustering at whatever values it happens to have, so you cannot attribute any effect to one property. Generating the data also means I know the true row count for every query.

How do you know the results are not just noise?

Every query runs seven times, the first is discarded, and I report the median with its spread. Nineteen automated checks run before any measurement is accepted, including validating my generated data against PostgreSQL's own internal statistics.

Did you find anything you got wrong yourself?

Yes, three measurement errors, and they are all reported in the paper rather than removed. One of them was an explanation I had to withdraw after testing my own claim and finding it did not hold.

If you run short of time

Slides 8, 9, 10 and 13 are the ones that matter. Slide 7 and slide 11 can be summarised in one sentence each and skipped. Do not skip slide 13; the recommendations are what the audience takes away.